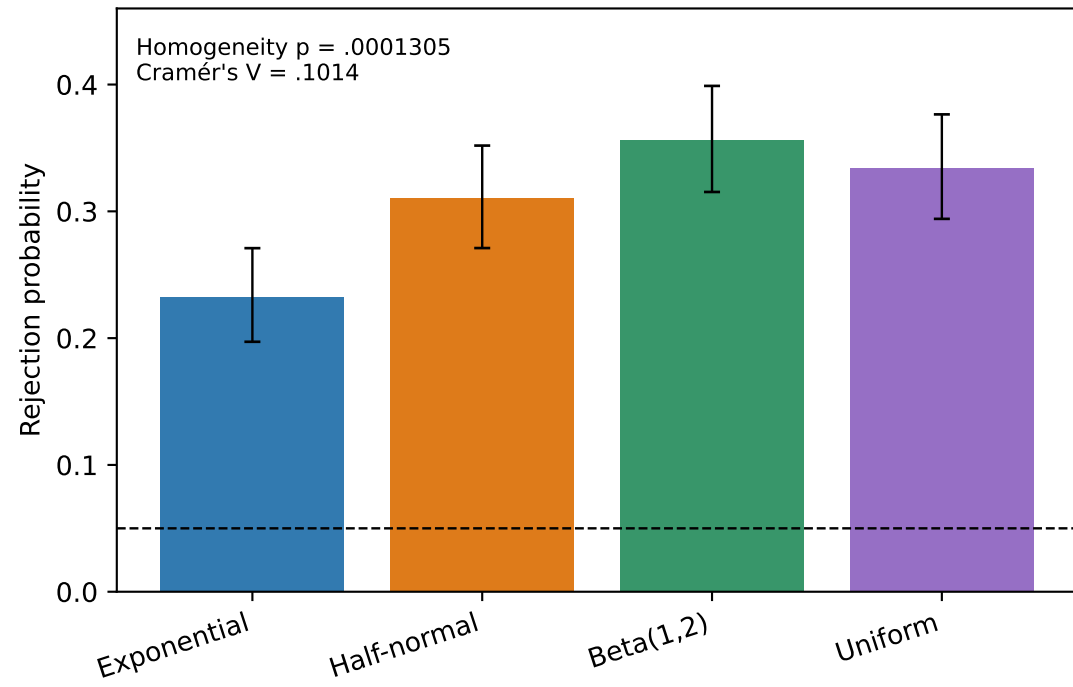


Figure 3. Family residuals unexplained by the index: studentized-permutation evidence only
A. Legacy families at matched first-order index ($I_n \approx 0.445$)



B. Prospective fifth-family residual (all predictions lie above observed rates)

